

Smart Toll AI Gate – Project Explanation

Good morning sir/madam,

Today I am going to explain my project titled **“Smart Toll AI Gate.”**

The main aim of my project is to automate toll booth operations using Artificial Intelligence and reduce traffic congestion, waiting time, and manual work at toll gates. I got this idea from real-world traffic problems where manual verification takes more time and creates long vehicle queues at toll booths.

In my project, the operator captures the vehicle image using a webcam. The image is then sent from the React and TypeScript frontend to the Node.js and Express backend. After that, I used Gemini Vision AI to analyze the vehicle details such as vehicle type and number plate from the captured image. Based on the detected vehicle type, my system automatically calculates the toll amount.

For developing my project, I used React and TypeScript for the frontend interface, Node.js and Express for backend processing, and Firebase Firestore for real-time database management. I used Gemini Vision AI because it can analyze complete vehicle details from images, unlike traditional OCR systems which mainly focus only on text recognition.

In my project, Firebase Firestore stores the transaction details and provides real-time synchronization without refreshing the application. I also implemented secure backend API handling to protect sensitive API keys and improve application security.

The major advantages of my project are reduced manual work, faster toll processing, real-time data management, and improved traffic flow at toll booths. In the future, I can improve this project by adding automatic payment systems, RFID integration, and smart city traffic management features.

Overall, my project shows how AI technology can be used to improve transportation systems

and automate real-world toll management processes.

Thank you.